DOMAĆA ZADAĆA BR $\boxed{3}$

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$\boxed{8.1}$ $n = 3$.

POTPORNI VEKTORI: | SKALIRANO:

$\vec{x}_1 = (2, -5, 15)$, $y_1 = -1$, $\alpha_1 = 0.5$ | $\vec{x}_{1s} = (-3.25, -3, -7.083)$

$\vec{x}_2 = (1, 8, 305)$, $y_2 = -1$, $\alpha_2 = 0.8$ | $\vec{x}_{2s} = (-3.5, 10, -11.75)$

$\vec{x}_3 = (1, -6, 225)$, $y_3 = 1$, $\alpha_3 = 0.4$ | $\vec{x}_{3s} = (-3.5, -4, 10.42)$

w_0 SE POMIČE

$\mu = (15.2, 1000)$

$\sigma = (4.1, 12)$

NOVI PRIMJER

$\vec{x} = (1, 2, -20) \rightarrow$ SKALIRANO: $\vec{x}_s = (-1.5, 4, -10)$

NAS ZANIMA: BEZ SKALIRANJA; SA SKALIRANJEM

$$h(\vec{x}) - h(\vec{x}_s) = \sum_{i=1}^{n} \alpha_i y_i \vec{x}^T \vec{x}^{(i)} + w_0 - \sum_{i=1}^{n} \alpha_i y_i \vec{x}_s^T \vec{x}^{(i)} - w_0$$

$$= \vec{w}^T \vec{x} - \vec{w}^T \vec{x}_s = \boxed{-373.20}$$

$$\vec{w} = \sum_{i=1}^{n} \alpha_i y_i \vec{x}_s^{(i)} = \begin{bmatrix} 1.275 \\ -10.1 \\ 39.474 \end{bmatrix}$$

Rj: $\boxed{D}$

$\boxed{9.1}$

$\alpha = 0 \Rightarrow$ NIJE SUPORNI VEKTOR

$$D = \{(\vec{x}^{(i)}, y^{(i)})\} = \{((1, -3, 6), y = -1), \cancel{((4, -4, 1), -1)}, ((-2, 4, 1), 1)\}$$

$C = 0.001 \Rightarrow \lambda = \frac{1}{C} = 1000$

$\vec{\alpha} = (0.001, 0, 0.001)$; $w_0 = -0.943$

$\boxed{R \doteq 1.9585}$ $\boxed{B}$

$$E_R(\vec{w} \mid D) = ?$$

$$E(\vec{w} \mid D) = \sum_{i=1}^{N} \max(0, 1 - y^{(i)} h(\vec{x}_i)) + \frac{\lambda}{2} \|\vec{w}\|_2^2$$

BEZ w_0

$$= (3 \cdot 10^{-3} + 0 + 1.914) + \frac{1000}{2} \cdot 8.3 \cdot 10^{-5}$$

$$\vec{w} = \sum_{i=1}^{n} \alpha_i y_i \vec{x}^{(i)} = -0.001 \cdot \begin{bmatrix} 1 \\ -3 \\ 6 \end{bmatrix} + 0.001 \begin{bmatrix} -2 \\ 4 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \cdot 10^{-3} \\ 3 \cdot 10^{-3} \\ -6 \cdot 10^{-3} \end{bmatrix} + \begin{bmatrix} -2 \cdot 10^{-3} \\ 4 \cdot 10^{-3} \\ 1 \cdot 10^{-3} \end{bmatrix} = \begin{bmatrix} -3 \cdot 10^{-3} \\ 7 \cdot 10^{-3} \\ -5 \cdot 10^{-3} \end{bmatrix}$$

$h(\vec{x}_1) = \vec{w}^T x_1 + w_0 = -0.997$

$h(\vec{x}_2) = \vec{w}^T x_2 + w_0 = -1.003$

$h(\vec{x}_3) = \vec{w}^T x_3 + w_0 = -0.914$

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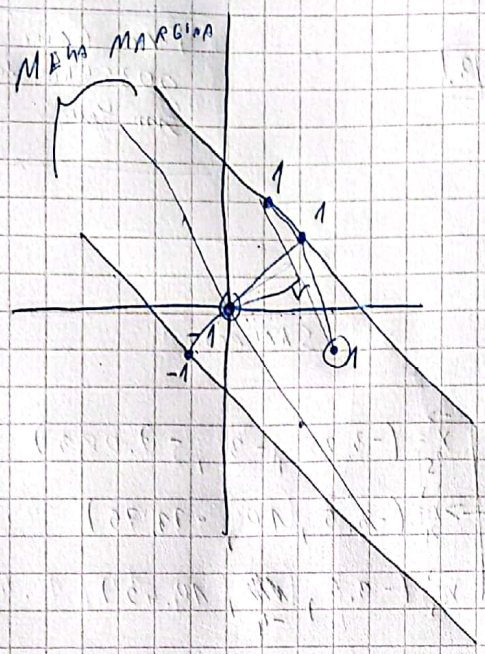
9.2



① TVRDA MARGINA

$$M_1 = \frac{\sqrt{2^2+1^2}}{2} = \frac{\sqrt{5}}{2}$$

② MEGA MARGINA:

$\alpha_1 = 1,\ \alpha_2 > 0,\ \alpha_3 > 0,\ \alpha_4 > 0$

$\alpha_5 = 1$

U MARGINI

$C = 1$ $\quad M_2 = \frac{3\sqrt{2}}{2}$

$$\boxed{\frac{M_2}{M_1} = \frac{\frac{3\sqrt{2}}{2}}{\frac{\sqrt{5}}{2}} = \frac{3\sqrt{2}}{\sqrt{5}} = \frac{3\sqrt{10}}{5}}$$

$\boxed{B}$

Rj.

10.1 $N = 5$

$\vec{y} = (1, 1, -1, -1, 1)$

JEZGRENA MATRICA K SA SLIKE

$C = 1$

$\vec{\alpha} = (\overset{\alpha_1}{0}, \overset{\alpha_2}{0.754}, \overset{\alpha_3}{0.754}, \overset{\alpha_4}{1}, \overset{\alpha_5}{1})$

NIJE SUPORT VEKTOR

NA MARGINI

SUPORT ili U MARGINI

$L(y_4, f(x_4)) = ?$

$L(y_4, f(\vec{x}_4)) = \max(0, 1 - y^{(4)} \cdot f(\vec{x}_4))$

$\boxed{= 0.0269}$

Rj: $\boxed{A}$

$$f(\vec{x}) = \sum_{i=1}^{N} \alpha_i y_i K(x_i, \vec{x}) + w_0$$

$f(\vec{x}_4) = \alpha_2 \cdot y_2 \cdot K(x_2, \vec{x}_4)$

$+ \alpha_3 \cdot y_3 \cdot K(\vec{x}_3, \vec{x}_4)$

$+ \alpha_4 \cdot y_4 \cdot 1$

$+ \alpha_5 \cdot y_5 \cdot K(x_5, x_4) + w_0$

$= -1.16425 + 0.1906$

$= -0.973626$

$w_0 = ?$

$y \cdot f(x) = 1$ $\quad$ za $0 < \alpha < 1$

$1 \cdot \left(\sum_{i=1}^{N} \alpha_i y^i K(x_i, x_1) + w_0\right) = 1$

$w_0 = 1 - \sum_{i=1}^{N} \alpha_i y^i K(x_i, x_1)$

$w_0 = 0.1906$

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10.2

$\mathcal{H}_{C,\gamma}$

PODNAĆENJE PRENAUČEN

$C = \{2^{-5}, 2^{-4}, 2^{-3}, 2^{-2}, 2^{-1}, 2^{0}, 2^{1}, 2^{2}, 2^{3}, 2^{4}, 2^{5}\}$

$\gamma = \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}, 10^{2}, 10^{3}, 10^{4}, 10^{5}\}$

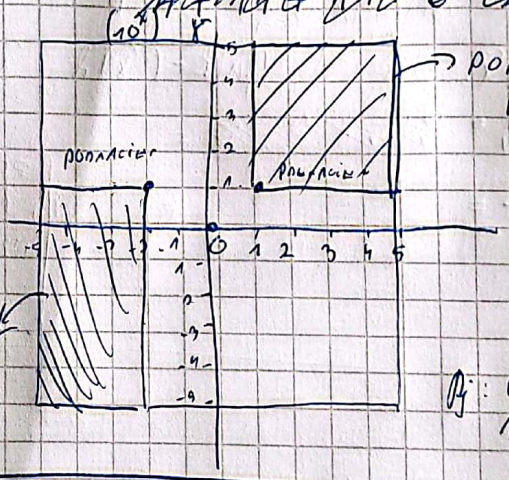
~~PODNAUČENI MODEL~~

Što je $C \uparrow$ kompleksnog modela je veća

Što je $\gamma \uparrow$, to su primjeri manje slični tj. model može biti skoro prenaučenost → složeniji model.

VRIJEDI IOŠ IZRADITI:

~~[illegible]~~



→ PODNAĆENI I NAIH
$5 \times 5 = 25$

ZA SVE OSTALE NEMOŽEMO REĆI SIGURNO JER TE POTKRIJEPE NIJE BILO OPTIMALNO

PODNAĆENI NAIH:
$4 \cdot 7 = 28$

tj. UKUPNO
$11 \cdot 11 - 25 - 28 = \boxed{68}$ $\boxed{A}$

11.1

$D = \{(\overset{x_1}{\text{"WATER"}}, 0), (\overset{x_2}{\text{"VODA"}}, 1), (\overset{x_3}{\text{"ZRAK"}}, 1), (\overset{x_4}{\text{"LUFT"}}, 2), (\overset{x_5}{\text{"FEUER"}}, 2)\}$

LJ SE MOŽE PREPOZNATI NAO SLOVO U NE NAO ZRAKA, ALI ISPADNE ISTO

$k(x_1, x_2) = \frac{|x_1 \cap x_2|}{|x_1 \cup x_2|}$

$x = $ "ZEMLJA"

$k(x_1, x) = \frac{2}{9}$ | $k(x_2, x) = \frac{1}{9}$

$k(x_3, x) = \frac{1}{8}$ | $k(x_4, x) = \frac{1}{9}$

$k(x_5, x) = \frac{1}{9}$

3NN

UZIMA: x_3 (1)
x_4 (0)
x_2 (1)

⇓

VRAĆA NEKO

$\boxed{A}$

Težinski 3-NN

za klasu 0: $\frac{2}{9} = 0.22$

za klasu 1: 0.361

za klasu 2: $\frac{2}{9} = 0.222$

↓ VRAĆA KLASU 1

Ajde ajde: $\boxed{B}$ 1.1

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